

Understand the problem & data

The dataset is team statistics for the 32 teams in the NFL for the year 2024 and includes attributes such as...

Offensive Statistics

- FGM: # of field goals made in regular season
- Pass_TD: # of passing touchdowns in regular season
- Rush_TD: # of rushing touchdowns in the regular season

Defensive Statistics

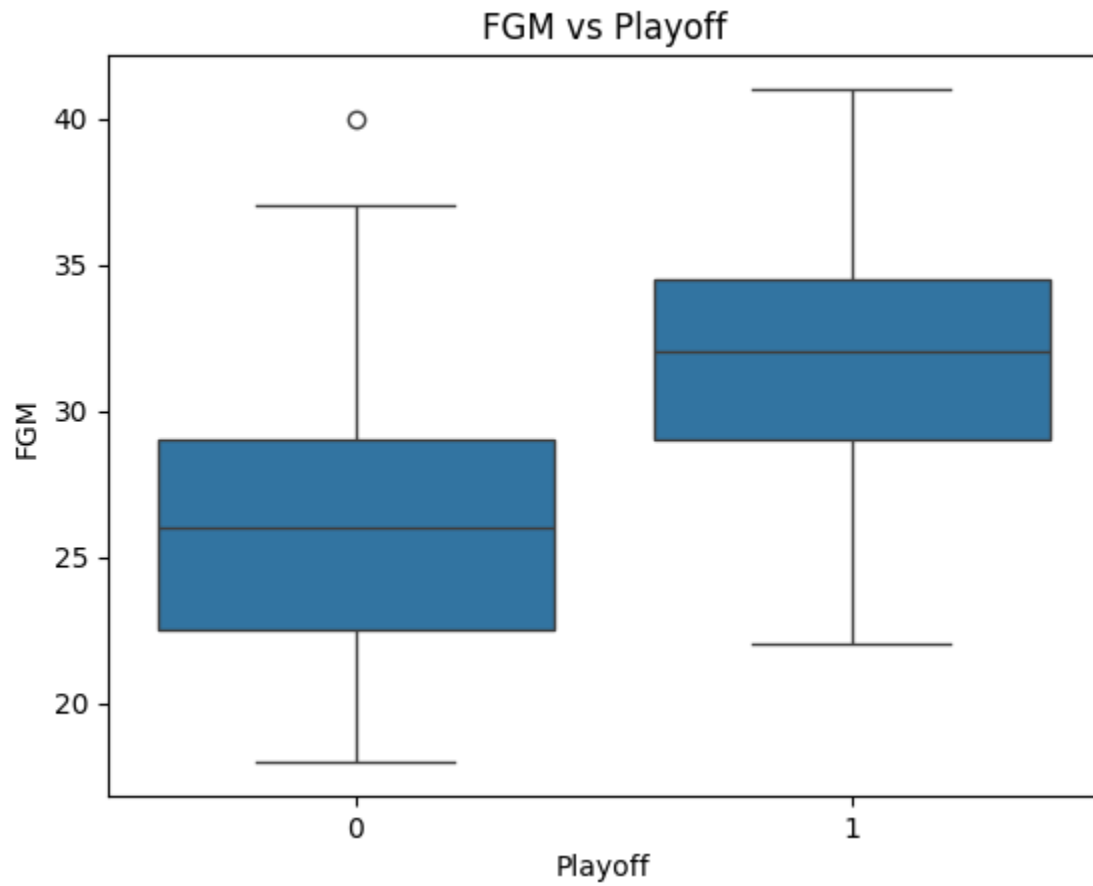
- INT: # of interception for the defense of the team in regular season
- FF: # of forced fumbles for the defense of the team in the regular season
- Sck: # of sacks for the defense of the team in the regular season

There is also a classification, Playoff, where 1 = the team made playoffs and 0 = if the team didn't make playoffs.

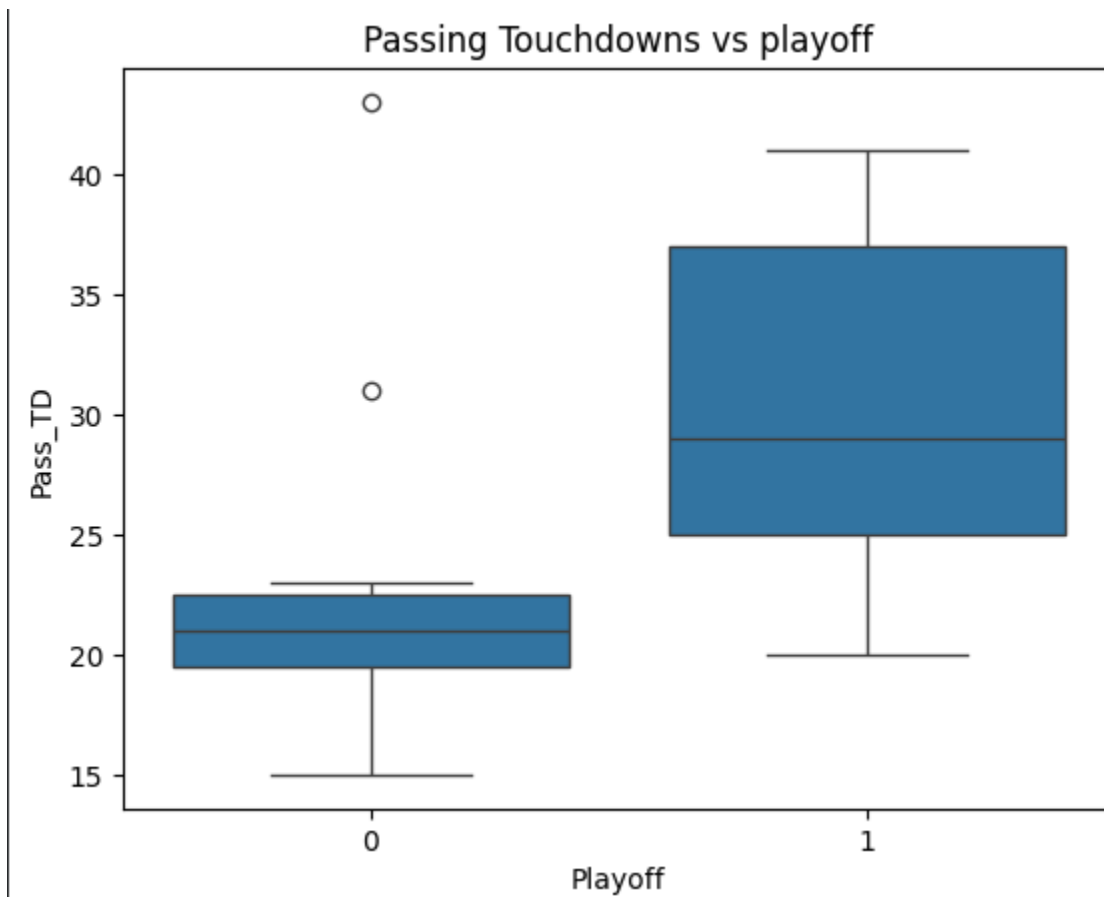
We split the dataset into a training (80%) and test set (20%). Our training set has 26 teams, 11 of which have made playoffs. Our test set has 6 teams, 2 of which have made playoffs.

The goal of statistical analysis is to analyze the performance of the Kansas City Chiefs in the 2024 season relative to other teams in the NFL, specifically related to whether or not teams made the playoffs.

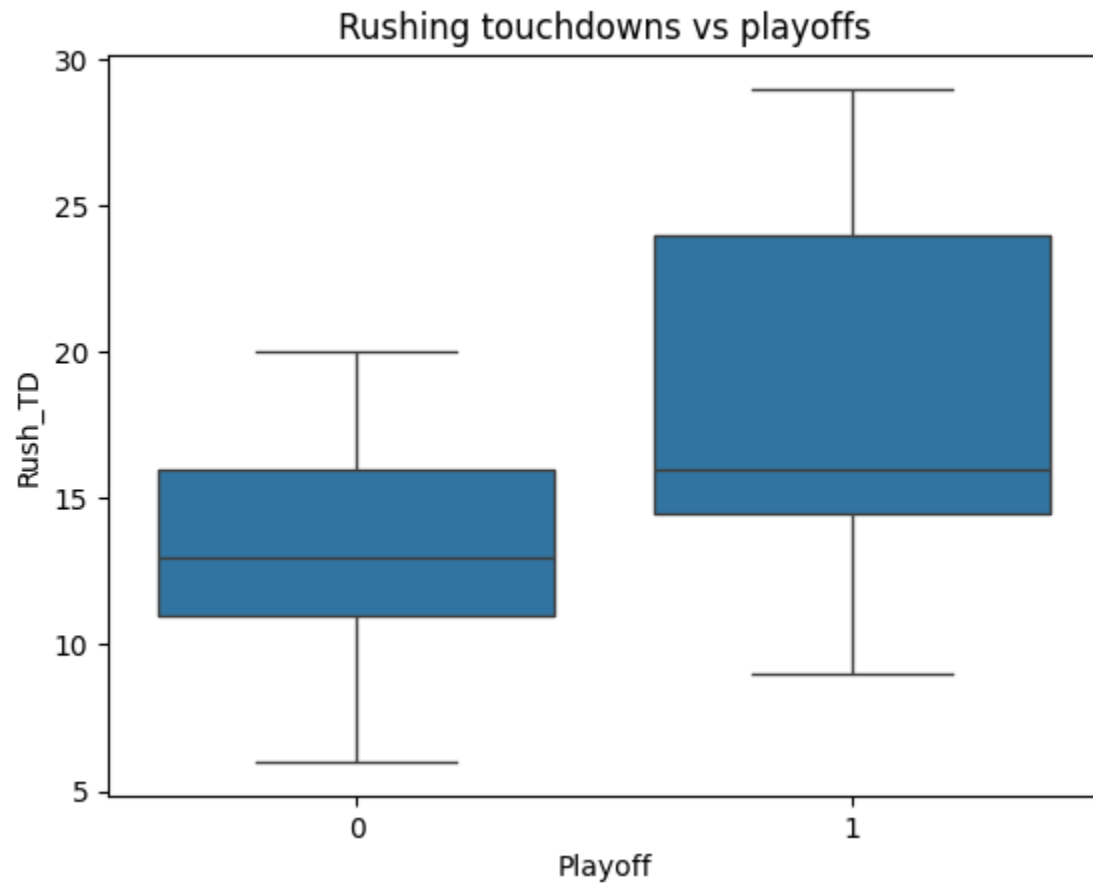
For the sake of simplification, all of the following plots for exploratory analysis will be boxplots, since we are interested in teams that did or did not make playoffs, and that variable is categorized as either 0 or 1.



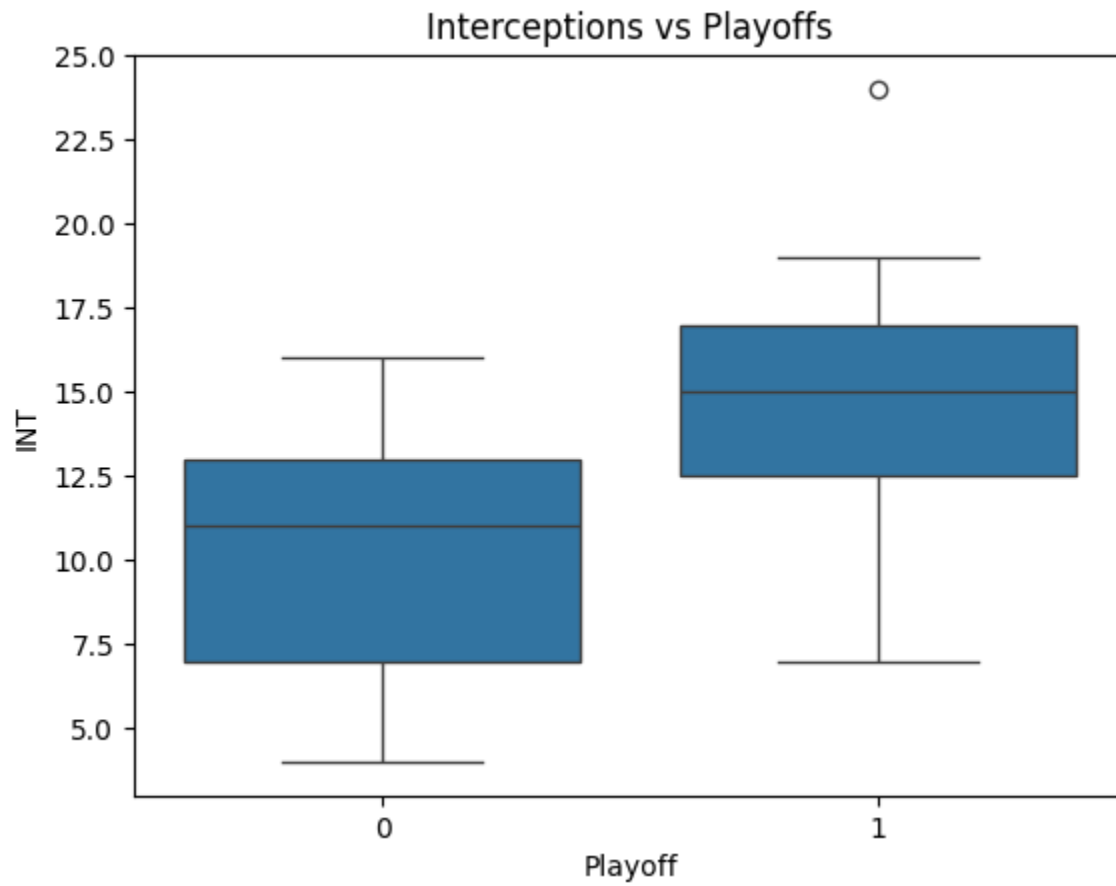
On average, teams that make playoffs tend to have a higher number of field goals made during the regular season than teams that didn't.



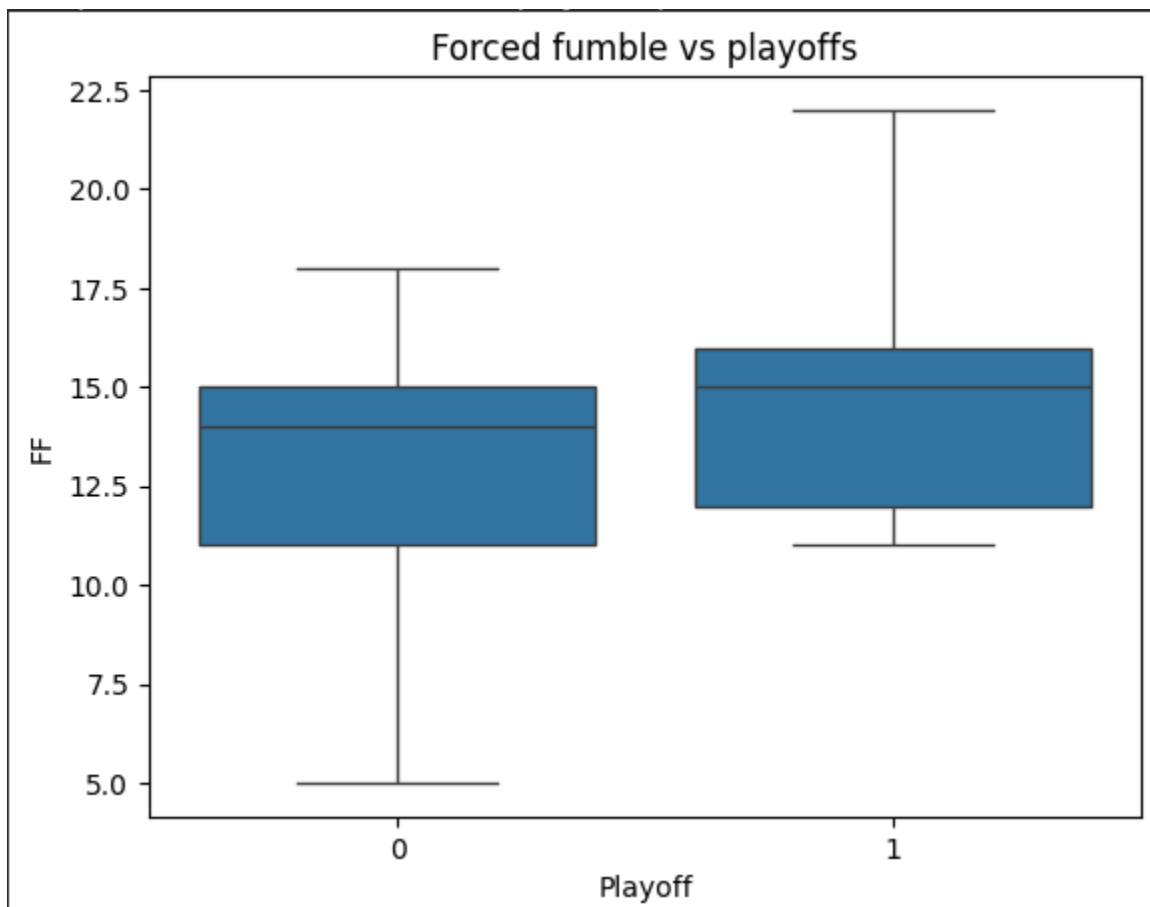
On average, teams that make playoffs tend to have a higher number of passing touchdowns than teams that didn't during the regular season. We do see 2 outliers for teams that have a high number of passing touchdowns but didn't make playoffs.



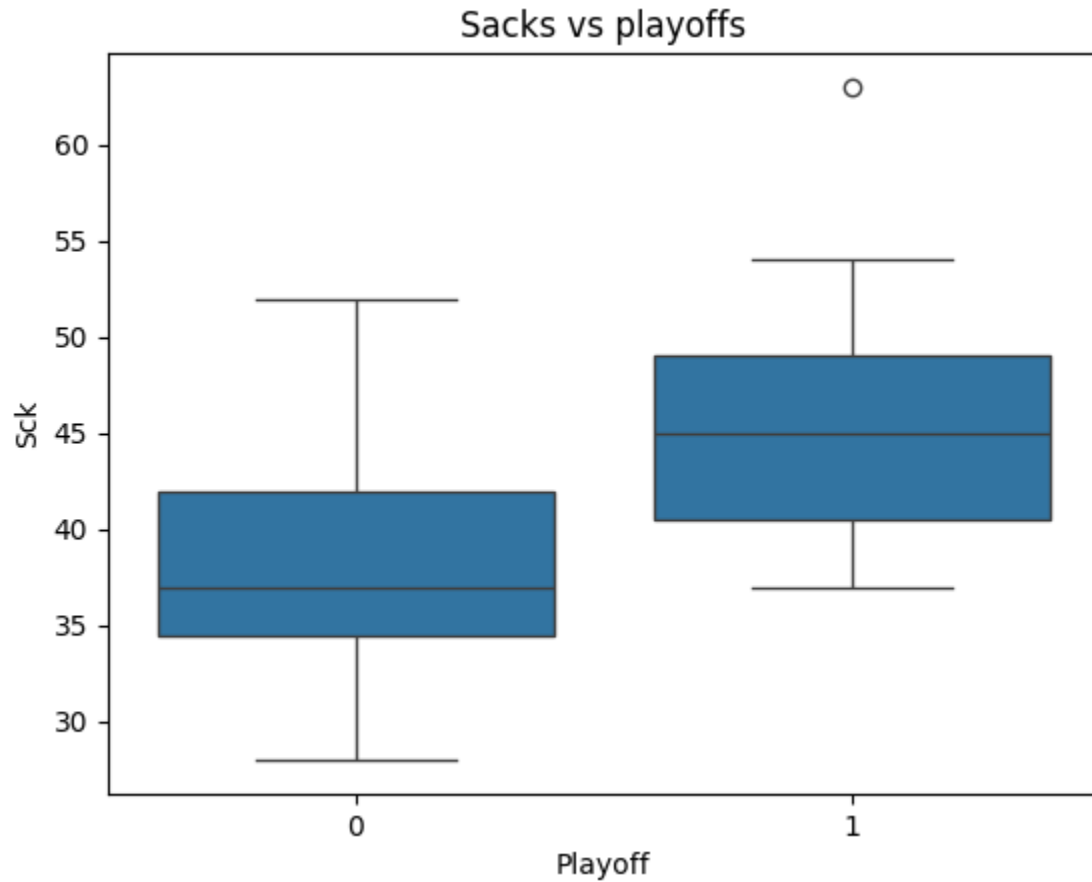
On average, teams that make playoffs tend to have a higher number of rushing touchdowns than teams that didn't during the regular season.



On average, teams that made the playoffs had a higher number of interceptions than teams that didn't, during the regular season.



As shown in the graph above, there does not appear to be a significant relationship between teams that did or didn't make playoffs depending on the number of forced fumbles they had during the regular season.



On average, teams that made playoffs tend to have a higher number of sacks compared to teams that didn't, during the regular season.

To detect any issues with multicollinearity we created a correlation matrix to see how each independent variable correlates to each other.

	FGM	Pass_TD	Rush_TD	FF	INT	Sck	Playoff
FGM	1.000000	-0.041424	0.010186	0.254827	0.514503	0.217960	0.393592
Pass_TD	-0.041424	1.000000	0.222080	0.013511	0.269767	0.317525	0.492240
Rush_TD	0.010186	0.222080	1.000000	0.260857	0.149313	-0.060951	0.500667
FF	0.254827	0.013511	0.260857	1.000000	0.188420	0.177697	0.297090
INT	0.514503	0.269767	0.149313	0.188420	1.000000	0.303502	0.467191
Sck	0.217960	0.317525	-0.060951	0.177697	0.303502	1.000000	0.501634
Playoff	0.393592	0.492240	0.500667	0.297090	0.467191	0.501634	1.000000

As shown above, the highest value of correlation between any two variables is between INT and FGM, however, the value is 0.515 so we can assume that the correlation between the two variables isn't significant enough to raise concerns about multicollinearity.

Model Specification

If all team statistics (offensive and defensive) were to be used to fit a support vector machine, we would have a 6-dimensional predictor profile, so the support vector classifier looks for a 5-dimensional hyperlane to separate the data.

We look at the following Kernels and parameters to fit a support vector machine

Kernel	Parameters
Linear	C
Polynomial	C, degree,
RBF (Radial basis function)	C, Gamma

Model Estimation

When tuning a support vector machine using 5-fold cross-validation for a linear kernel, we get...

C = 0.1

Teams that are incorrectly classified in the training set for this model would be...

- Chiefs

When tuning a support vector machine using 5-fold cross-validation for a polynomial kernel, we get...

C = 0.1

Degree = 2

Teams that are incorrectly classified in the training set for this model would be...

- Chiefs
- Cowboys

When tuning a support vector machine using 5-fold cross-validation for a radial basis function, we get...

$C = 10$

Gamma - 0.01

There were no teams that were incorrectly classified using the RBF support vector machine

Model Assessment

Predicted	0	1
Actual		
0	3	0
1	1	2

This is the classification matrix for the linear kernel support vector machine. As we can see, there is only one misclassification.

Predicted	0	1
Actual		
0	3	0
1	1	2

This is the classification matrix for the polynomial kernel support vector machine. As we can see, there is one misclassification.

Predicted	0	1
Actual		
0	3	0
1	1	2

This is the classification matrix for the RBF kernel support vector machine. As we can see, there is one misclassification. With this in mind, a RBF support vector machine would be best to proceed with when considering all predictor variables, since even though it did the same on the test data compared to the other kernels, it did the best when classifying teams in the training set.

Model Specification

A 6-dimensional space is difficult to interpret, so for the KC Chiefs, we aim to prepare an analysis for the offensive and defensive coordinators. Here, we will highlight how the Chiefs performed relative to all other teams, using the entire dataset without any training/test splits to conduct our analysis. We will use a linear kernel where each model

has a two dimensional predictor space. This means that for the offensive coordinator, we will consider the following three predictor profiles: [FGM and Pass_TD], [FGM and Rush_TD], & [Pass_TD and Rush_TD]. For the defensive coordinator, we will consider the following three predictor profiles: [FF and INT], [FF and Sck], & [INT and Sck].

Model Estimation

Offensive Statistics

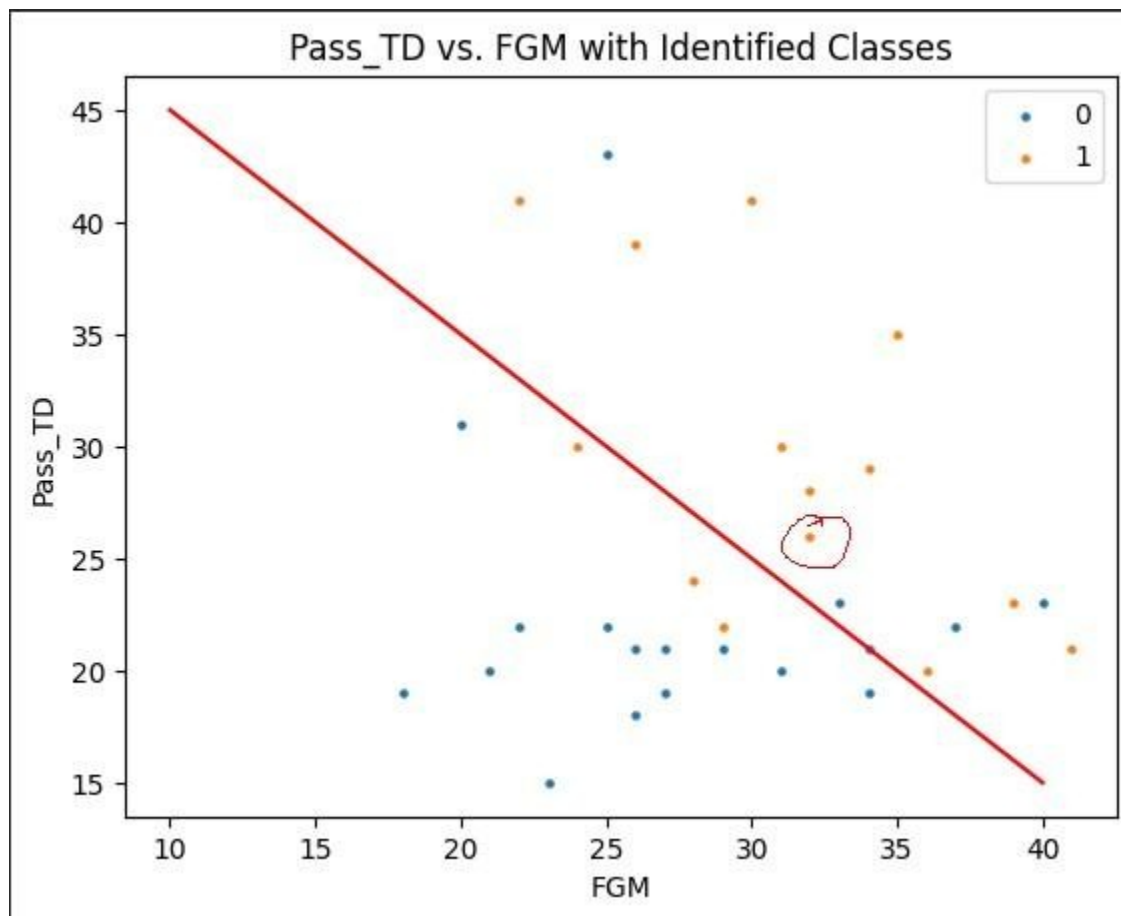
Support Vector Machine	Parameters	Estimated Line
[FGM and Pass_TD]	$C = 0.01$	$Y = -1x + 55$
[FGM and Rush_TD]	$C = 0.1$	$Y = -0.857x + 44.27$
[Pass_TD and Rush_TD]	$C = 0.01$	$Y = -0.801x + 38.934$

Defensive Statistics

[FF and INT]	$C = 0.01$	$Y = -0.375x + 17.692$
[FF and Sck]	$C = 1$	$Y = -0.751x + 52.258$
[INT and Sck].	$C = 0.01$	$Y = -1.8x + 64.801$

Model Deployment

Offensive coordinator analysis

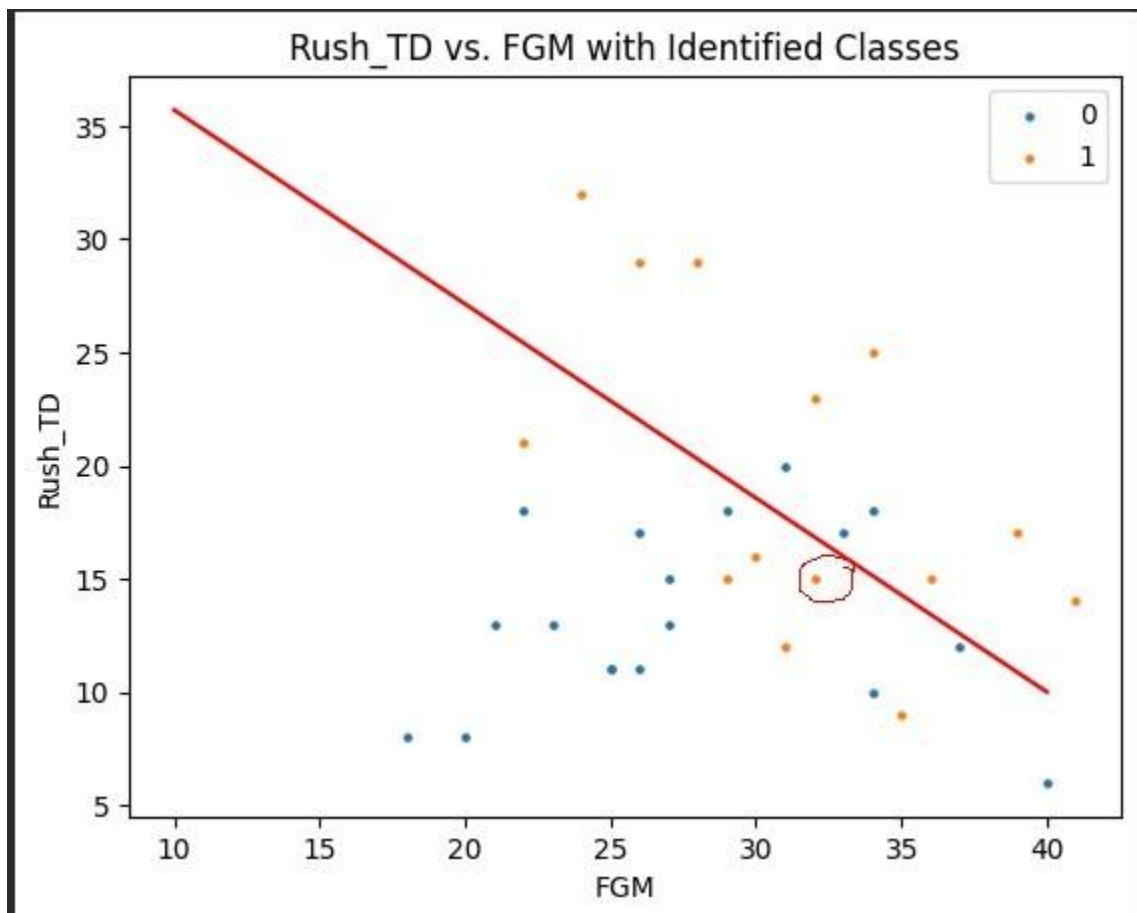


Estimated Line: $Y = -1x + 55$

Orange dots reveal a team that has made the playoffs, and blue represents teams that have not. There are 3 teams that made the playoffs but are incorrectly classified as not making the playoffs. The Kansas City Chiefs are circled in red.

An offensive coordinator should realize that passing touchdowns in the regular season are a much better way to predict whether or not a team makes the playoffs compared to the number of field goals made in the regular season.

Conclusion Remark: The chiefs had a relatively low number of passing touchdowns compared to other teams that made play offs, so maybe more or less attention can be placed on the number of field goals they make, and more on how to get more passing touchdowns.

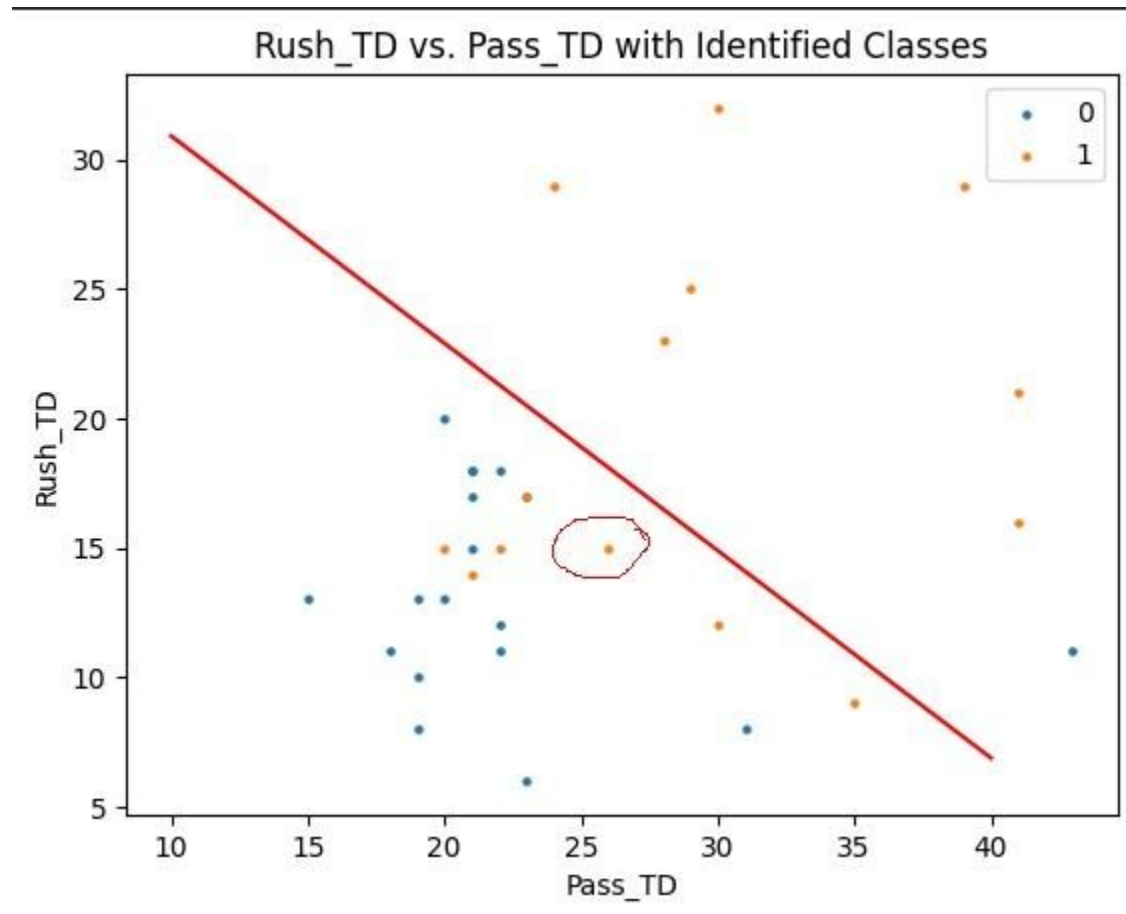


Estimated Line: $Y = -0.857x + 44.27$

There is no clear hyperplane that separates teams that did and did not make the playoffs in the 2024 regular season. There are 6 teams that made the playoffs but are incorrectly classified as not making the playoffs, including the chiefs, which are circled in red.

The offensive coordinator should learn that Rushing Touchdowns are a better predictor of whether or not a team makes the playoffs compared to the number of field goals they make during the regular season.

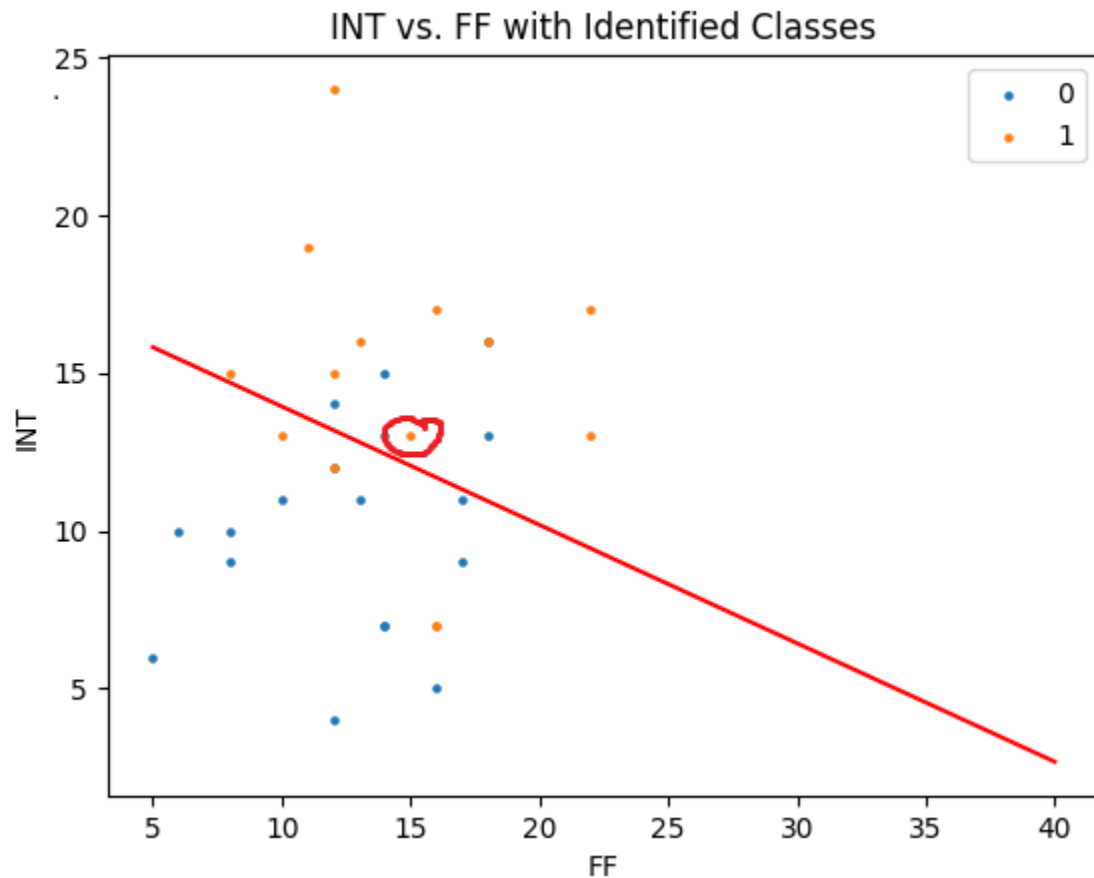
Concluding Remark: The Kansas City Chiefs had a low number of rushing touchdowns compared to other teams that made the playoffs, so focusing on rushing touchdowns during their practice can be beneficial.



There is no clear hyperplane that separates teams that did or didn't make the playoffs during the 2024 season. There are 7 teams that made the playoffs but are incorrectly classified as not making the playoffs, including the Chiefs, which are circled in red.

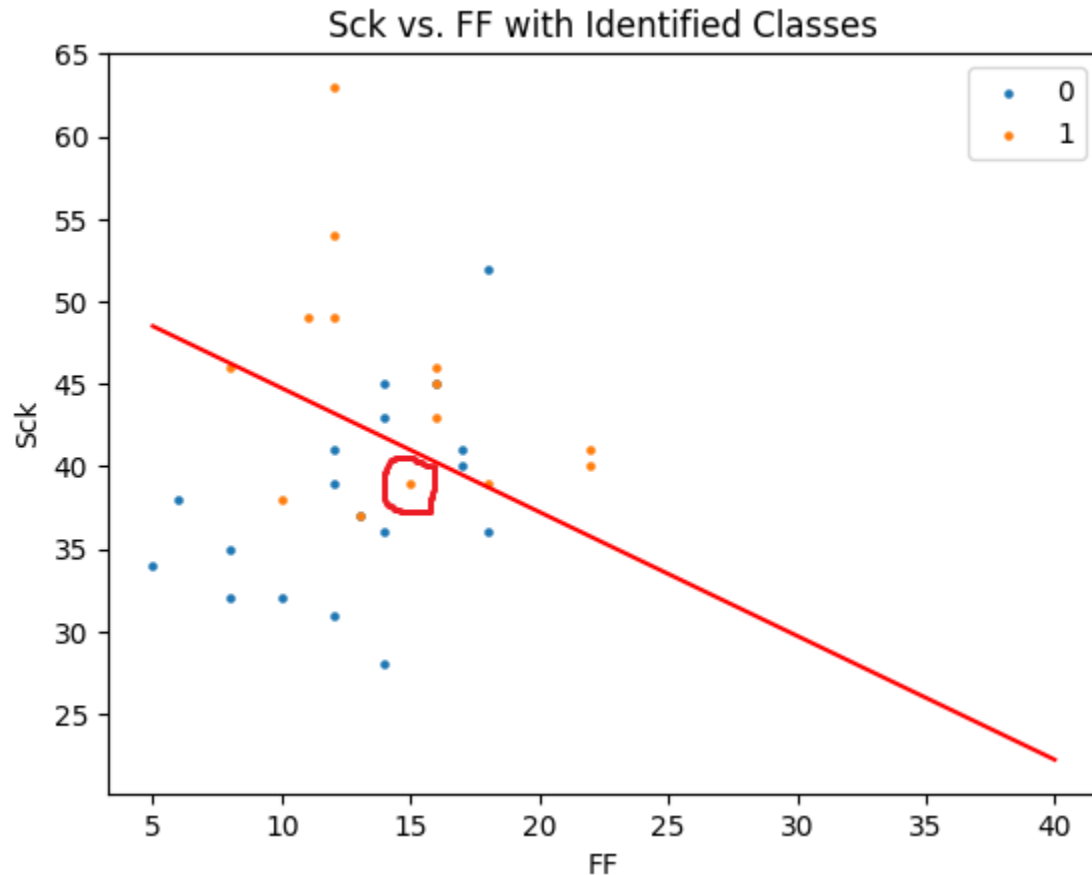
An offensive coordinator can learn that rushing touchdowns are a better predictor for if a team makes the playoffs or not compared to the number of passing touchdowns in their regular season.

Concluding remark: The Chiefs had a low number of rushing touchdowns compared to teams that made the playoffs. The chiefs did about average for the number of their passing touchdowns, so that is also an area that can be improved as well.



There is no clear hyperplane that separates teams that did or didn't make the playoffs during the 2024 season. There are 3 teams that made the playoffs but are incorrectly classified as not making the playoffs, not including the Chiefs, which are circled in red. A defensive coordinator can learn that the number of interceptions is a better predictor of whether or not a team makes the playoffs compared to the number of forced fumbles the defense of a team has in the regular 2024 season.

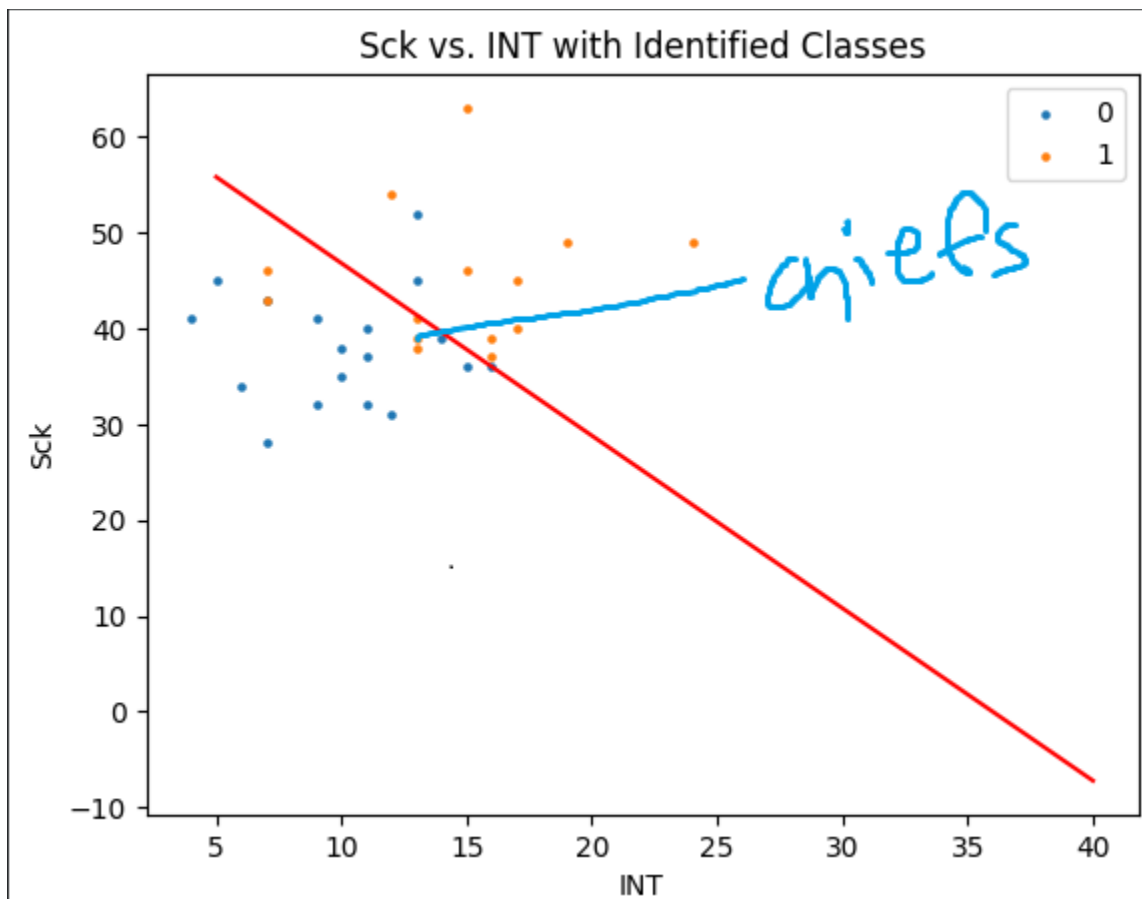
Concluding remark: The Chiefs have a low number of interceptions compared to other teams that made the playoffs, so interception practice should be utilized.



There is no clear hyperplane that separates teams that did or didn't make the playoffs during the 2024 season. There are 4 teams that made the playoffs but are incorrectly classified as not making the playoffs, including the Chiefs, which are circled in red.

A defensive coordinator can learn that the number of sacks is a better predictor of whether or not a team makes the playoffs compared to the number of forced fumbles the defense of a team has in the regular 2024 season.

Concluding remark: The Chiefs have a low number of sacks compared to other teams that made the playoffs, so the defensive coordinator should put more attention into their defense's ability to sack.



There is no clear hyperplane that separates teams that did or didn't make the playoffs during the 2024 season. There are 5 teams that made the playoffs but are incorrectly classified as not making the playoffs, including the Chiefs, which I have pointed out in blue.

A defensive coordinator can learn that the number of interceptions is a better predictor of whether or not a team makes the playoffs compared to the number of sacks the defense of a team has in the regular 2024 season.

Concluding remark: The Chiefs have a somewhat low number of interceptions compared to other teams that made the playoffs, so the defensive coordinator should put more attention into their defense's ability to intercept the ball from opposing teams.

Generative AI Usage Statement

I have not used Generative AI tools to prepare this submission